

Chapter 3: Regularization and Variable Selection in R

Syntax Notes, Exercises, and Worked Solutions

STAT452 – Applied Statistics for Engineers and Scientists II

Contents

1	Core ideas	1
2	R syntax	1
3	Exercises	1
4	Worked solutions	2

1 Core ideas

OLS can be unstable under multicollinearity or when many predictors are available. Ridge minimizes $SSE + \lambda \sum_j \beta_j^2$ and shrinks coefficients toward zero. LASSO minimizes $SSE + \lambda \sum_j |\beta_j|$ and can set coefficients exactly to zero. Predictors should usually be standardized before penalization. Selection should be evaluated by cross-validation or information criteria, not p-values alone.

2 R syntax

```
library(MASS)
ridge <- lm.ridge(mpg ~ wt + hp + qsec + disp + drat + gear,
                  data=cars, lambda=seq(0,30,length.out=100))
select <- which.min(ridge$GCV)
full <- lm(mpg ~ wt + hp + qsec + disp + drat + gear, data=cars)
step(full, direction="both", trace=0)
```

3 Exercises

Exercise 1: OLS baseline. Fit a full model and inspect correlations among predictors. Which coefficients might be unstable?

Exercise 2: Ridge path. Fit ridge over a grid of λ , plot coefficient paths, and choose λ by GCV. Explain the bias-variance trade-off.

Exercise 3: LASSO. Use the supplied coordinate-descent function to plot LASSO paths. Identify coefficients that become zero first and explain standardization.

Exercise 4: Information criteria. Compare stepwise selection using AIC and BIC. Why might formulas differ?

Exercise 5: Cross-validation. Compare small and full models using five-fold CV. Explain why lower training SSE is not enough.

4 Worked solutions

Solution 1. Strong correlations indicate overlapping information. OLS can be unbiased under the model, but coefficient variance and sensitivity can be large. $\square$

Solution 2. As λ grows, ridge coefficients move smoothly toward zero but generally do not equal zero. It trades a little bias for lower variance. $\square$

Solution 3. LASSO paths are piecewise and some coefficients hit zero. Standardization makes the penalty comparable across units. $\square$

Solution 4. AIC uses a lighter complexity penalty and often keeps more variables; BIC is more parsimonious. Both are heuristics, not proofs. $\square$

Solution 5. CV estimates performance on observations not used to fit each fold. A larger model may lower training error while increasing validation error through overfitting. $\square$

Checklist

standardize -> fit path -> tune lambda -> validate out of sample -> interpret sparsity.